

Pranesh S

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SUMMARY

AI/ML Engineer specializing in **Agentic AI** and Generative AI — B.Tech (AI & ML) graduate — **Immediate Joiner** — Building Production LLM systems: Agentic pipelines on **MCP** and **AWS Bedrock AgentCore**, RAG platforms, and **7B-parameter model fine-tuning**. Open-source maintainer: **100+ public repos, 270+ stars, 3 PyPI packages, 35+ live Hugging Face Spaces**.

TECHNICAL SKILLS

Generative AI & Agents: LLM fine-tuning (LoRA, QLoRA), RAG, embeddings & semantic search, quantization (4-bit, GGUF), AI Agents, Agentic AI, Multi-Agent Systems, MCP, LangChain, LangGraph, CrewAI, prompt engineering, Context Engineering, Harness Engineering

Machine Learning & Deep Learning: Transformers, PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, Deep Learning, NLP, computer vision (EfficientNetV2, MTCNN), model evaluation

Languages & Backend: Python, Java, SQL, Data Structures & Algorithms, System Design, Flask, FastAPI, REST APIs, PostgreSQL, ChromaDB

Cloud & MLOps: AWS Bedrock AgentCore, Azure AI Foundry, Modal, Docker, Hugging Face, PyPI, Git/GitHub (Version Control), Vercel, Render, Supabase

EXPERIENCE

- AI Engineering Intern — Wipro Connected Services (formerly Harman DTS)** Feb 2026 – Jun 2026
- Achieved **82% executable-SQL generation (37/45)** on the **Spider benchmark** by building an **NL2SQL2NL** system on the **Model Context Protocol (MCP)** with schema-aware prompting, enabling context-aware, multi-turn plain-English querying of relational databases for business users.
 - Reduced SQL composition time by **40%** and passed **100% of an 8-query business demo suite** by delivering the pipeline to non-technical stakeholders with conversational, **Matplotlib**-based chart generation.
 - Deployed team-built **Agent** to **AWS Bedrock AgentCore Runtime** by containerizing **FastAPI** services as **linux/arm64 Docker** images against the AgentCore service contract and publishing to **Amazon ECR**, covering both HTTP and MCP runtime protocols.
 - Achieved **92% tool-agent alignment accuracy** and **35% faster agent development** by building a **Multi-Agent Workflow Optimizer** with ontology-driven requirement parsing and **AgentBank** validation, preventing **5+ duplicate tool implementations**.

PROJECTS

- RagObserve — RAG Observability Platform** | *Python, FastAPI, SQLite, PyPI* github.com/Pranesh-2005/RagObserve
- Delivered end-to-end **RAG observability** — tracing retrievals, LLM calls, latency, and cost across providers — by building a **local-first platform** with **LangChain** and **LlamaIndex** integrations that runs fully offline with zero external dependencies.
- Deepfake Video Detector** | *Python, EfficientNetV2, MTCNN, Gradio* hf.co/spaces/PraneshJs/fakevideodetect
- Earned **27 GitHub stars** for an open-source deepfake detection system flagging AI-generated video via frame-by-frame analysis, built by combining **MTCNN** face detection with **EfficientNetV2** classification; shipped a live **Gradio** demo on Hugging Face with a user feedback loop.
- PromptGuard — Prompt-Injection Detection Model** | *Python, BERT, Transformers, PyPI* hf.co/PraneshJs/PromptGuard
- Reached **150+ downloads** on Hugging Face by training a **BERT text-classification model** on prompt-injection datasets to guard LLM apps against malicious inputs; also shipped as the **guardix** PyPI library.
- Persona Mimic Chatbot — Qwen2.5-7B + LoRA + RAG Memory** | *Python, Unsloth, Modal, GGUF, Gradio*
- Produced **14K instruction-tuning samples** from **120K+ raw WhatsApp messages** by engineering a parsing/dedup/burst-windowing pipeline; fine-tuned **Qwen2.5-7B-Instruct** with **LoRA (Unsloth)** on serverless **Modal A10G** GPUs.
 - Enabled long-term conversational memory by embedding **30K+ chat bursts** with multilingual-e5 for top-k **RAG** retrieval at chat time; exported **4-bit GGUF quantized** models for local inference and deployed a Gradio Space on Hugging Face.

ACHIEVEMENTS & OPEN SOURCE

- github-readme-stats-fast:** GitHub profile stats generator with a companion GitHub Action — **270+ stars, 59 forks**
- PyPI packages:** *guardix* (prompt-injection guard), *Text-Emotion* (text emotion detection), *DocTok* (document token counter)
- 35+ live Hugging Face Spaces** spanning RAG, agents, model-internals visualizers (GPT-2, ViT, YOLO), and AI tooling

EDUCATION

Bannari Amman Institute of Technology 2022 – Apr 2026
Bachelor of Technology (B.Tech) in Artificial Intelligence and Machine Learning — CGPA: 7.97/10

CERTIFICATIONS

- Anthropic:** Introduction to Model Context Protocol; Model Context Protocol: Advanced Topics
- Microsoft:** Azure AI Fundamentals (AI-901)
- freeCodeCamp:** Machine Learning with Python